

# Web-Based Indoor Navigation System

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**Abstract.** In multistorey buildings with varied shapes, such as a collection of hexagonal structures, it can be difficult for users to identify the correct path from one room to another. During peak hours, locating the right direction or a free pathway becomes even more challenging due to similar layouts. To overcome this and ensure a user-friendly navigation experience, this paper focuses on delivering an internal indoor navigation system with an accessible interface. By leveraging the detailed building layout and infrastructure, a digital indoor navigation system has been designed for easy use by any visitor. The system enables users to search for room numbers, explore floor layouts, and receive step-by-step indoor navigation guidance. With a simple interface, the system enhances accessibility and ensures smooth movement throughout the building. Through digital innovation, the solution improves time efficiency and supports a seamless navigation experience for all users.

**Keywords:** Indoor Navigation · Geospatial Data · Web Application · Graph Theory · Smart Campus

## 1 Introduction

Indoor navigation has emerged as a critical challenge in modern academic and urban infrastructures, particularly in large, multistorey buildings with complex layouts. The three-story building used in this work comprises 27 rooms per floor. Due to its scale and intricate design, it can be challenging for students, faculty, and visitors to find specific rooms, especially if they are not familiar with the environment.

While outdoor navigation has been effectively addressed using the Global Positioning System (GPS), satellite-based positioning systems perform poorly in indoor environments, as observed by P. Davidson [17]. Positioning accuracy is severely reduced indoors due to signal attenuation from building materials and the lack of direct line-of-sight to satellites. For indoor pathfinding applications, GPS-based solutions are therefore inappropriate.

As buildings become bigger and more complex, the need for reliable indoor navigation systems has naturally increased. Alternative positioning and navigation strategies are essential in indoor environments due to their special challenges, such as intricate interior layouts and dynamic infrastructure changes. As surveyed by P. Davidson and R. Piché [17], research on this indoor "last-mile" navigation issue is still in progress, but there are a lot of chances for advancements in indoor positioning and pathfinding technologies.

### 1.1 The Opportunity in Indoor Navigation

In large public venues like airports, hospitals, shopping centres, and university campuses, efficient indoor navigation is especially vital. In such environments, users frequently feel anxious and confused when navigating strange areas, which can result in missed deadlines and a bad user experience all around. From an operational standpoint, inadequate pathfinding can lead to bottlenecks, including more staff time spent providing directions and

underutilization of specific building spaces. The need for reliable and user-friendly indoor positioning and navigation systems is emphasized by ensuing economic and social costs linked to inefficient indoor navigation.

According to J. Schiller and A. Voisard [21], the market for indoor location-based services (iLBS) is therefore expected to grow significantly due to the growing uptake of smart building technologies and the need for improved user experiences. While earlier research on indoor pathfinding focuses on sensor-driven navigation ecosystems supported by mobile computing and data analytics, this paper advances the field by using algorithms on digital floor plans to calculate paths efficiently, following the approach of A. Kumar [5] and as explained by M. Youssef [11]. The suggested system provides a lightweight, scalable, and implementation-friendly substitute for sensor-dependent frameworks by incorporating the A\* algorithm for optimal pathfinding to enable quick, deterministic, and accurate route computation between indoor locations without relying on extensive sensor infrastructures, as discussed by H.A. Karimi [1].

### ***1.2 Paper Overview: The INDOORNAV Solution***

This work presents INDOORNAV, a web-based indoor navigation system designed to address the limitations of conventional indoor pathfinding solutions. By allowing users to visualize intricate floor plans and search for indoor locations, the technology simplifies navigation and increases user productivity. INDOORNAV was created especially for indoor locations, with a focus on room-level accuracy and smooth navigation across multiple floors, in contrast to conventional mapping applications.

The system is accessible through standard web browsers, eliminating the need for users to install dedicated mobile applications and lowering the barrier to adoption. On widely accessible devices like computers and cell phones, this design guarantees instant accessibility for users. While the frontend offers a clear and responsive interface that enables users to acquire navigation routes by providing source and destination room identifiers, the backend evaluates geometric and topological representations of indoor environments to calculate best paths between places.

### ***1.3 Paper Objectives and Goals***

Following the foundational framework established by H.A. Karimi [1], the primary aim of this paper is to develop a web-based indoor navigation system that delivers efficient and intuitive guidance within complex indoor environments. The system features a user-friendly interface that supports accurate route calculation, scalability, and easy access. To accomplish this, the following objectives are set:

- **Web-Based System Development:** Build a platform-independent web application accessible through standard web browsers. This eliminates the need for users to download or install native software, ensuring compatibility across various devices and operating systems.
- **Shortest Path Computation:** Implement the A\* pathfinding algorithm to determine optimal routes between indoor points such as classrooms, auditoriums, elevators, and staircases. This ensures effective and reliable navigation by minimizing travel distance and estimated traversal time within complex indoor layouts.
- **Visual Map Representation:** Provide graphical visualization of computed routes on digital floor maps to provide users with clear, understandable, and easy-to-follow navigation instructions.
- **Scalable System Architecture:** Following the guidance of A. Stepanov [19], design a modular and extensible system architecture capable of accommodating future

expansions, including additional buildings, floors, and functional enhancements, thereby ensuring long-term adaptability of the solution.

#### ***1.4 Novelty and Contribution***

The novelty of INDOORNAV is that it uses algorithms on digital floor layouts to determine the optimal indoor routes rather than sensors or specialized hardware. As noted by H.A. Karimi [1] and J. Wang [7], standard indoor navigation systems often rely on dense sensor networks, mobile beacons, or specialized hardware. In contrast this system computes optimal routes exclusively using algorithm techniques and spatial data. As noted by H.A. Karimi [1] and A. Kumar and R. Sharma [5], this makes the solution lightweight, cost-effective, and straightforward to deploy across academic institutions and large-scale indoor facilities.

INDOORNAV includes emergency-exit routing in addition to point-to-point navigation, explicitly modelling exits as priority destination nodes in the graph. Unlike traditional evacuation techniques that depend on real-time sensor data or infrastructure upgrades, this approach enables the system to dynamically calculate safe and fastest evacuation routes using the same graph-based framework, as detailed by X. Liu [3] and R. Müller [8].

The system is made to be accessible and scalable. As in J. Wang [7] and R. Müller [8], multi-level navigation is made possible by representing each floor as a distinct navigation graph that is connected by staircases and elevators. Unlike mobile-app-only solutions, the web-based interface reduces the adoption barrier by providing rapid access across devices without requiring installation (J. Schiller and A. Voisard [21]). GeoJSON, NetworkX, GeoPandas, Flask, and Matplotlib were used to create the open-source, modular framework, which also acts as a useful resource for future research and indoor navigation implementations.

Overall, the combination of algorithmic efficiency, integrated emergency routing, multi-floor scalability, and web-based accessibility distinguishes INDOORNAV from prior work (H.A. Karimi [1], A. Kumar [5], A. Verma [9], J. Schiller and A. Voisard [21]), offering a robust and reproducible framework for academic, commercial, and smart-building indoor navigation applications.

## **2 Literature Review and Theoretical Foundations**

The development of an indoor navigation system is predicated on extensive study in the fields of indoor positioning systems, algorithmic pathfinding, and spatial data modelling. Indoor navigation has emerged as a critical research domain due to the limitations of Global Positioning System (GPS) signals within enclosed environments. Large-scale facilities like hospitals, universities, and business complexes require precise and dependable navigation technology to help people navigate complex indoor environments.

Modern developments in geospatial information systems (GIS) and web-based technologies are combined with classical computer science foundations, especially graph theory and artificial intelligence. These interdisciplinary approaches enable robust, scalable, and user-friendly indoor navigation systems.

### ***2.1 Spatial Data Modelling and Graph Construction***

Accurate spatial data representation is a fundamental requirement for any effective indoor structured spatial modelling in enhancing routing accuracy and user experience. According to a 2024 study published by Taylor & Francis, led by Y. Zhang [2] and S. Zlatanova [12], precise spatial representations and three-dimensional mapping significantly improve interior route planning by providing more accurate distance computations and decision-point modelling. Similarly, a 2023 study presented by SCIRP on navigation systems, as emphasized by J. Lee [13], Prior research emphasizes the role of X. Liu [3], proposes a

network-based indoor navigation model using three-dimensional spatial graphs, where nodes represent rooms or intersections and edges represent corridors or pathways, facilitating efficient shortest-path computation.

Building upon these principles, the system models indoor spaces using GeoJSON, an open-standard format designed for encoding geographic data structures, as standardised by the Open Geospatial Consortium [15]. As highlighted by Pointr [4] and the Open Geospatial Consortium [15], GeoJSON is lightweight, interoperable, and well-suited for modern web-based QGIS applications. It enables precise definition and manipulation by representing indoor features such as rooms, corridors, staircases, and entry points as geometric entities including polygons and line strings. The suitability of GeoJSON for indoor navigation and location-based services has been emphasized in recent industry studies by Pointr [4], and Open Geospatial Consortium [15].

To process and analyze this spatial data, the system employs Python-based geospatial and graph-processing libraries. GeoPandas is an extension of conventional tabular data analysis that allows geometrical operations to be performed on different objects, whereas NetworkX is a framework to build and analyze graph-based networks in detail. The study by A. Kumar in IJACSA [5] emphasizes the efficiency of the combination of these tools when it comes to spatial data management of an indoors environment, graph visualization, and calculations related to navigation.

The transformation of a physical building layout into a navigable graph is a critical step for algorithmic pathfinding. The first step of this process is to digitize architectural floor plans with the help of GIS tools like QGIS. The structural components such as rooms, corridors, doorways, staircases, elevators, etc. are followed as a polygonal or a linear element, which records the important decision points in the interior space, as detailed by D.A. Schilling [14].

The spatial data in digitized form is then subject to computing geometric libraries like Shapely, which represent spatial features as formal geometric objects and allow the identification of adjacency relationships. Intersections, shared boundaries, and walkable connections are extracted and converted into a graph structure using NetworkX, where access points form nodes and navigable paths form edges. The pipeline of converting spatial to graph representation is a transformation of detailed floor plans of the interior into abstract graphs that can be used to compute a path with a high level of efficiency by relying on the already developed pathfinding algorithms.

## ***2.2 Pathfinding Algorithms: Dijkstra vs. A\****

Once the indoor environment is represented as a graph, a pathfinding algorithm is required to determine the most efficient route between a source and a destination. According to R. Sedgewick and K. Wayne [20], S.M. LaValle [6] and D.A. Schilling et al. [14], A\* and Dijkstra's algorithm are among the most widely adopted and foundational shortest-path techniques used in network-based navigation systems.

The algorithm of Dijkstra is deterministic and ensures the minimum optimal path by searching all possible nodes in a systematic search order in terms of path cost. Although its reliability is well-founded, the algorithm may be computationally infeasible in large or dense graphs, as it evaluates all possible paths without regard to the direction of the destination. This exhaustive search approach can create unnecessary computational costs in complicated indoor settings.

In contrast, according to S.M. LaValle [6] and R. Sedgewick and K. Wayne [20], the A\* algorithm improves efficiency by incorporating a heuristic function that estimates the remaining cost from a given node to the target. This heuristic significantly reduces the number of nodes that are traversed by enabling the algorithm to prioritize nodes that have a

high probability of leading to the destination. Geometric heuristics like Euclidean distance work especially well in the structured indoor environment, like corridors and the fixed building layout, as described by A. Millonig [16]. As a result, A\* typically offers faster execution times while still guaranteeing optimality when an admissible heuristic is used, as discussed by R. Sedgewick [20].

Given these characteristics, A\* achieves a favourable balance between computational efficiency, accuracy, and resource usage, making it especially suitable for indoor navigation applications, according to studies in indoor navigation by H.A. Karimi [1], A. Kumar and R. Sharma [5], and J. Wang et al. [7]. A comprehensive discussion of the theoretical foundations and performance trade-offs of these algorithms is provided by LaValle in Planning Algorithms [6].

### ***2.3 Multi-Floor Navigation***

Efficient indoor navigation requires effective handling of vertical movement between floors. According to J. Wang [7], previous study highlights the conceptualization of staircases and elevators as the graphical connections to offer smooth travel between floors of the building. Similarly, graph-based multi-floor navigation frameworks proposed in earlier studies demonstrate the effectiveness of node-based vertical connectivity for continuous path computation, as described by R. Müller [8].

In this approach, each floor is represented as an independent navigation graph. Staircases and elevators are modelled as special connector nodes that link corresponding locations across floors. For each vertical connector, a pair of nodes is created—one on each connected floor—and an edge is introduced with a weight representing traversal cost such as time or effort, allowing dynamic adjustments if conditions change, as described by M. Worboys [18].

### ***2.4 Future Directions: AR and AI Integration***

While the current system concentrates on accurate and efficient graph-based navigation, future enhancements could improve user interaction and situational awareness. One potential extension is the integration of Augmented Reality (AR) to visualize navigation cues within the user's physical environment, an approach explored by Verma [9] in recent indoor navigation studies.

Lightweight AI techniques like context-aware guidance or voice-based destination input could be implemented to enhance usability. These enhancements are intended to be complementary features rather than necessary components in order to guarantee that the system is dependable, efficient, and operating even in the lack of sophisticated hardware or sensor infrastructure.

## **3 System Architecture and Design**

The INDOORNAV system is designed to ensure outstanding scalability, maintainability, and clarity. It is a high-level, three-tier structure which effectively enables data handling and processing from user input to final path visualization.

### ***3.1 High-Level Three-Tier Architecture***

The application is divided into logical layers using a three-tier architecture, wherein each layer is designated with a set of different tasks.

Presentation Layer (Frontend): HTML5, CSS3, and JavaScript are used to provide an interactive visual layer to the user. The page allows users to enter their source and destination rooms. To retrieve the navigation path, the REST API exchanges information between the

frontend and backend, which is then presented to the user. Overall, this layer oversees every aspect of the user interface and user experience.

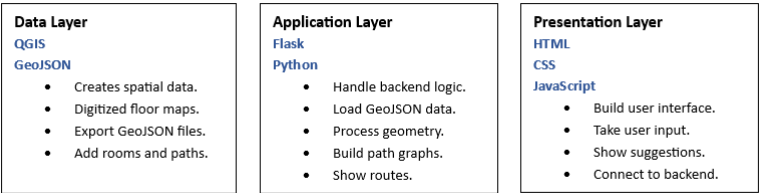


Fig. 1. The Three-Tier Architecture of the Indoor Navigation System.

Application Layer (Backend): Python and Flask web framework are used to create the core of the system which handles all the backend logics. Pathfinding is executed by this layer which takes in requests from the frontend and then returns the results back. Hence, it acts as a mediator between the user's requests and the underlying data.

Data Layer: This layer stores and manages the spatial data of the building. The floor plans are contained in a GeoJSON file. The geometric and semantic data from this layer is used to build the navigation graph. In reference with the building layout and structure of the building, preserving and retrieving of all information is overseen.

Fig. 1 illustrates the three-tier architecture of the indoor navigation system.

3.2 Detailed System Design Flow

Within the application layer, the processing logic is further broken down into a layered design flow. This flow details the sequence of operations that transform raw spatial data into a navigable path.

1. User Interface Layer: This is an interactive layer. Users can input their source and destination room numbers. Over here the information is verified and then sent to the backend. And finally, display the finished map.
2. Data Layer: The GeoJSON building map, which has the spatial representation of floors, rooms, and corridors, serves as the system's foundation. This data has been loaded for processing.
3. Processing Layer: Complex topological and geometric analysis is handled by this layer. The GeoJSON data is parsed, room and corridor geometries are extracted, and important features like room centroids and spatial relationships are calculated using libraries like GeoPandas and Shapely. Graph construction is then done using this processed data.
4. Pathfinding Layer: This layer determines the best course of action. Using NetworkX, a graph is created, with nodes standing in for rooms or corridor intersections and edges for paths that can be navigated. A\* algorithm is applied to the graph for finding the optimal path between the user's start node and end node.
5. Visualization Layer: This layer creates an eye-catching visual representation after calculating the shortest path. The system creates a clear, highlighted map by superimposing the calculated path onto the floor plan using Matplotlib.
6. System Output: The highlighted navigation map is returned to the user interface as a high-quality PNG image for instant and useful display.

Detailed system design and processing flow is depicted in Fig. 2.

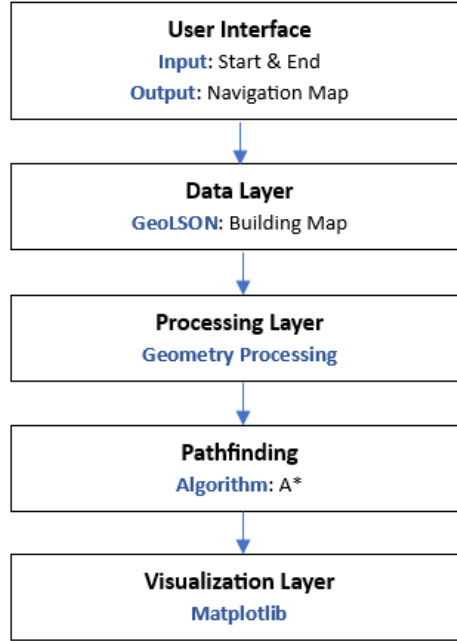


Fig. 2. Detailed system design and processing flow.

### 3.3 End-to-End Data Flow

The entire path of a user request through the system, from the first interaction to the last result displayed, is depicted by the end-to-end data flow. Each element will work flawlessly to provide a prompt and precise response thanks to this smooth flow.

The steps are as follows:

1. User Input: Start and end destination rooms.
2. Request to Backend: Through a REST API call, the frontend transmits this data to the Flask backend.
3. Data Processing: The backend constructs a navigation graph from indoor spatial data and applies the A\* algorithm to compute the shortest path. Data implementation includes QGIS-based digitization, Flask-powered route computation using GeoPandas and NetworkX, and real-time visualization through an HTML, CSS, and JavaScript frontend.
4. Path Visualization: The computed path is plotted onto the relevant floor plan using Matplotlib.
5. Frontend Response: The generated map is sent back to the frontend by the backend, in the form of a high-quality PNG image.
6. User Output: The user receives a clear and attractive navigation path image.

Fig. 3 shows the sequence of operations from user input to path display.

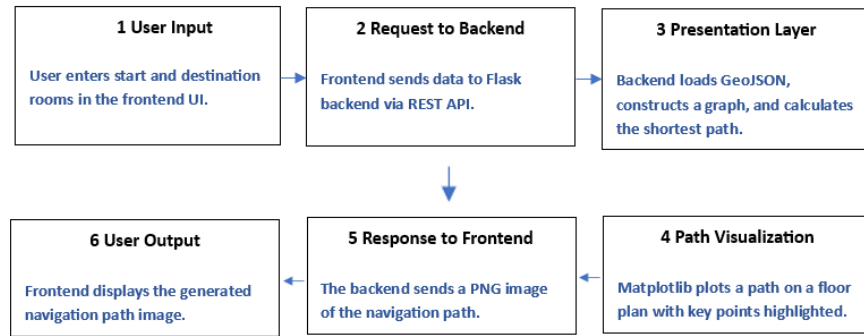


Fig. 3. The sequence of operations from user input to path display.

## 4 Methodology and Implementation

The successful deployment of the INDOORNAV system resulted from a structured, multi-phase process encompassing data collection, data preparation, and the development of frontend and backend components. The technology stack was carefully selected. These are widely recognized for their effectiveness in geospatial analysis, web development, and data visualization. This section offers a thorough analysis of the technologies employed as well as the particulars of each system component's implementation. The approach emphasizes modularity, reusability, and clear documentation in accordance with accepted software engineering practices.

### 4.1 Technology Stack

The system was developed using open-source and industry-standard technologies. These were chosen based on their reliability, compatibility, and support for spatial data processing. Spatial data was prepared using Quantum Geographic Information System (QGIS) from architectural blueprints.

The spatial data was stored and exchanged using Geographic JavaScript Object Notation (GeoJSON).

The backend was implemented in Python using the Flask web framework to provide RESTful navigation services. Data handling and geometric computations were handled via GeoPandas and Shapely. Construction and analysis of the indoor navigation graph was done using NetworkX. Matplotlib supported visualization of calculated paths on the building floor plans.

HTML5, CSS3, and JavaScript are used to build the frontend interface. This enables user interaction and asynchronous communication with the backend through API calls.

### 4.2 Implementation Details

Preparing the spatial data, developing the backend logic, and creating the frontend interface were the three primary stages of the implementation.

#### 4.2.1 Spatial Data Preparation

High-resolution images of the architectural blueprints were imported into QGIS (Quantum Geographic Information System). PNG (Portable Network Graphics) images were chosen because they preserve clarity and scale better than PDF files during processing. Each floor plan was aligned to a local scale system to ensure that distances and measurements inside the building were accurate. Using QGIS, the building's main features—including walls, corridors, rooms, staircases, and entry points—were carefully traced to create precise digital maps that could be used for navigation and analysis.

Next, the definition of features was done. Each floor was modelled as a separate map layer to allow easy floor-by-floor analysis. Within each layer, stairs, corridors, and lifts were represented as polygons to capture their full shape, and rooms were also drawn as polygons to reflect their exact boundaries. This method ensures that the spatial relationships between different areas are correctly maintained, which is important for building the navigation routes.

Finally, the maps were exported as GeoJSON files. This format preserves both the shapes and the properties of each area, making it easy for the backend to process the data. The structure of the GeoJSON files followed standards like the Indoor Mapping Data Format (IMDF), ensuring compatibility and reliability for indoor navigation applications. Although the current implementation utilizes these GeoJSON outputs directly within the front-end interface, the same data can be persistently stored in a backend database or file system with minimal modification. This provides a clear pathway for future extensions involving long-term storage, data versioning, and multi-building scalability.

#### *4.2.2 Backend Logic*

The backend of the navigation system is built using Flask, which provides a web service that receives the starting and destination room numbers from the user. When a request is sent, the backend quickly processes the information and begins calculating the navigation route.

Upon receiving a request, the backend proceeds to load the relevant floor maps from GeoJSON files through GeoPandas. It calculates the centre points of the start and end rooms and aligns, or "snaps," these points to the nearest corridor. This ensures that navigation starts and ends on actual walkable paths rather than in the middle of rooms.

Using NetworkX, the system then constructs a network graph representing the building layout. Room entrances and corridor intersections serve as nodes, while the corridors connecting them serve as edges. Each edge is assigned a length based on the actual distance, which allows the system to accurately determine the shortest path.

According to graph-based indoor navigation approaches proposed by A. Kumar and R. Sharma [5] and X. Liu and J. Sun [3], the A\* algorithm is applied to a graph representation of the environment to determine the most efficient route from the starting point to the destination. The algorithm produces an ordered list of coordinates representing the path that a user should follow.

The backend uses Matplotlib to show the route so users can see the path immediately. By sending the image directly to the frontend without saving it to disk, the system avoids delays and keeps the process simple, fast, and easy to use on any device.

#### *4.2.3 Frontend Interface*

According to studies on indoor navigation systems by H. A. Karimi [1], A. Verma and S. Patel [9], and A. Millonig and G. Schechtner [16], the frontend of INDOORNAV provides a simple and intuitive interface for users to interact with the navigation system. Users can enter their starting and destination room numbers, which includes two text fields for input. Basic checks using HTML attributes ensure that the fields are not left empty before the request is sent.

The visual appearance and layout of the form provides a clean, responsive, and user-friendly design. CSS ensures that the input fields, buttons, and map display area are clearly organized and easy to use on different devices.

When the user submits the form, a JavaScript function is triggered to send the room information to the backend in a structured format. The backend processes the request and calculates the navigation path. The resulting map is returned as an image and displayed dynamically on the web page. This dynamic update, powered by JavaScript, allows the user

to see the route immediately without reloading the entire page. It provides instant visual feedback, making the navigation experience smooth, interactive, and easy to follow.

## **5 Results and Discussion**

INDOORNAV has been built out into a fully working application that validates what the paper was proposing methodologically. A visual output is provided for the shortest path. Routing accuracy came out to approximately 96–98% across the tests that were run.

The results confirm that the tech stack and overall architecture chosen for this project make a solid indoor navigation solution that is both effective and scalable. Looking at the key performance parameters like accuracy, speed, and general usability, route computation averaged under 200 ms and operational efficiency cleared 90%, both of which were targets of the paper.

A comparative analysis between shortest-path algorithms further strengthens these findings. Dijkstra's algorithm recorded a total computation time of 239 ms (189 ms processing + 50 ms overhead), whereas the A\* algorithm achieved a lower total time of 204 ms (189 ms processing + 15 ms overhead). This demonstrates that A\* is more efficient due to its heuristic-based approach, which reduces unnecessary node exploration and improves overall performance in indoor navigation scenarios.

### ***5.1 Strengths of the Proposed System***

A major strength of the INDOORNAV system is that it provides dynamic indoor navigation, which replaces user reliance on static maps or fixed signboards, as highlighted by J. Hightower [10]. Rather than requiring users to manually interpret directions, the system calculates the shortest path between two rooms and displays it clearly on a map of the building. In large or complex buildings where disorientation is common, this form of direct visual guidance helps in reducing confusion, minimising unnecessary time spent navigating, and providing a more structured and efficient experience overall.

The system also has the feature of emergency exit routing which proves to be crucial for user safety. In an emergency situation that requires evacuation, such as a fire, a user can simply enter "exit" and receive accurate and instant directions to the nearest emergency exit, limiting human error and optimising the evacuation process. The system instantly recalculates the shortest available route based on the layout which will allow for faster evacuations. Furthermore it can assist during emergency drills by providing clear, straightforward directions. INDOORNAV works by taking detailed indoor floor plans and transforming them into graph-based layouts, which allows the system to provide accessible routes. One notable feature is its ability to factor in outdoor pathways on the ground floor whenever these turn out to be quicker than going through the building itself — cutting down travel time and making the whole experience more efficient compared to users finding their way around manually.

The system holds up well as demands grow, and it adapts without much friction. Each floor gets modeled on its own before being connected to the others in a logical structure, meaning that bringing a new floor or an entirely separate building into the mix simply comes down to digitising the layout and plugging it in. This kind of modular setup means INDOORNAV can handle buildings of all shapes, sizes, and levels of complexity without needing to be rebuilt from scratch.

Unlike applications such as Google Maps which have been tailored for outdoor navigation, INDOORNAV follows a different approach. It runs purely on indoor spatial data, which

means it stays reliable even in environments where GPS is completely absent. That makes it a strong fit for places like hospital complexes, university campuses, academic buildings, and large commercial spaces where conventional navigation tools tend to struggle.

From a technical standpoint, the decision to build the system using an open-source technology stack has advantages of its own. The libraries used in the building of this system are open source, which cuts down the time needed to develop and troubleshoot while also sidestepping the costs that typically come with proprietary software licenses. The end result is a system that is built in such a way that it can actually be sustained and scaled over time without becoming a financial burden.

5.2 Input and Output



Start Room:

End Room:

Find Path

Fig. 4. INDOORNAV User Interface.

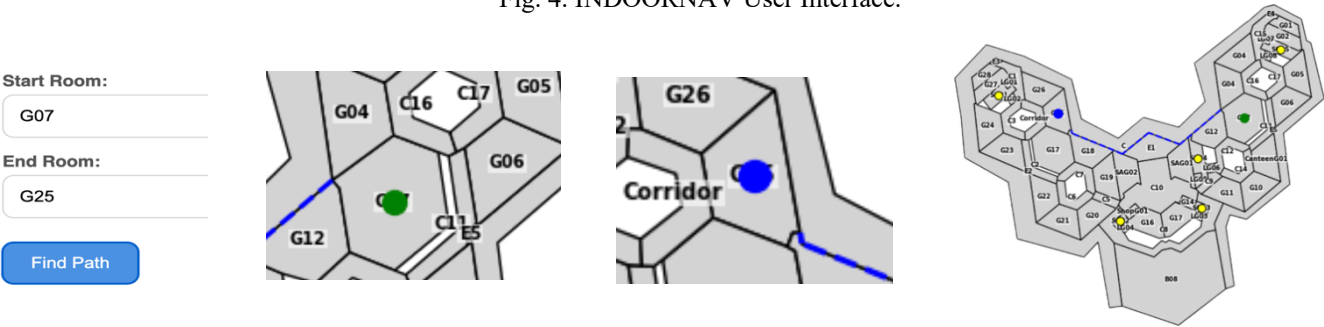


Fig. 5. Output for single-floor input navigation (Green dot – start point (G07), blue dot – end point (G25), yellow dots – staircases).

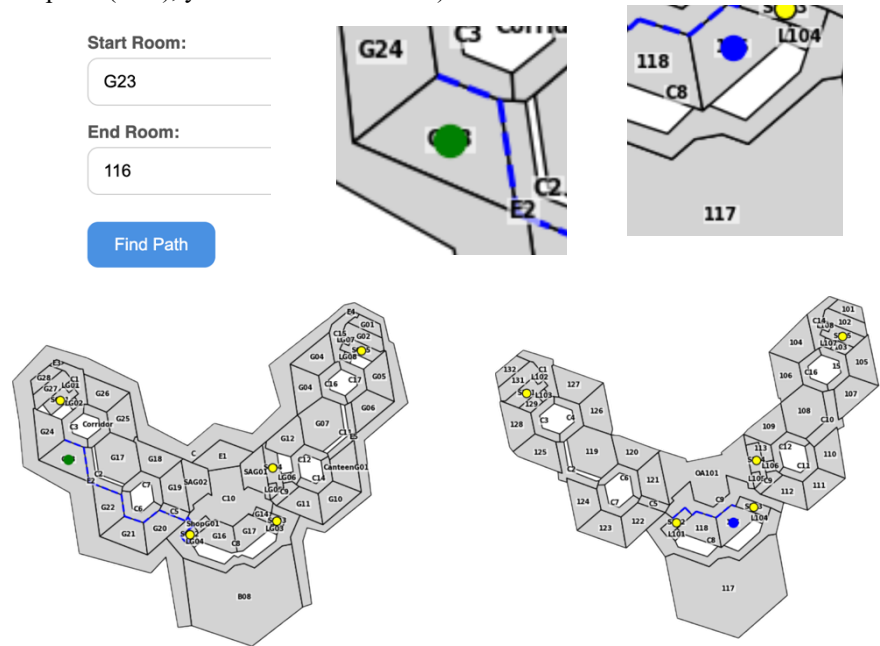




Fig. 6. Outputs for double-floor input navigation (Green dot – start point (G23), blue dot – end point (116), yellow dots – staircases, blue line – path).

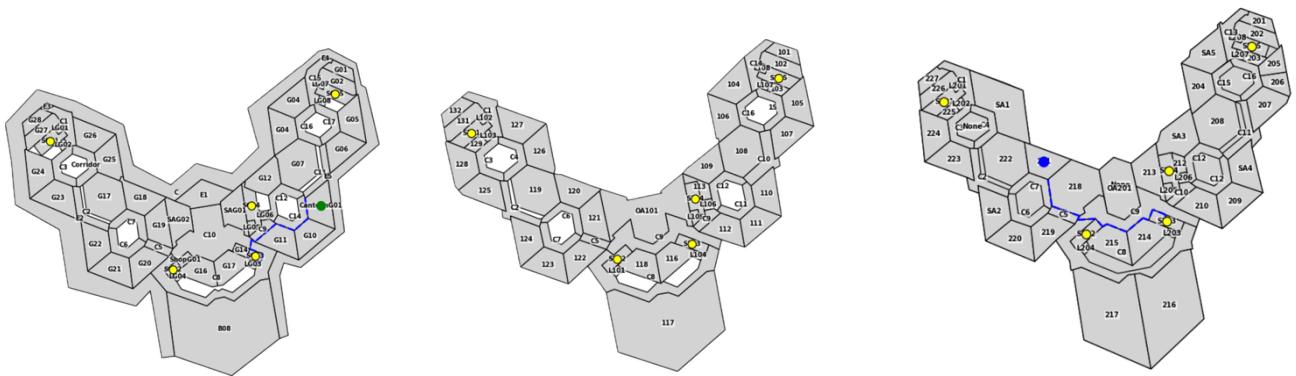


Fig. 7. Outputs for triple-floor input navigation (Green dot – start point (Canteen), blue dot – end point (221), yellow dots – staircases, blue line – path).



Fig. 8. Outputs for emergency exit navigation (Green dot – start point (G10), blue dot – end point (exit – E5), yellow dots – staircases, blue line – path).

### 5.3 Potential Applications

The core technology and methodology developed for this paper have broad applicability. The system could be used for any indoor environment.

Within a large hospital, the system can be used to navigate between different wards, labs, and clinics. In an airport, the system can assist users by facilitating terminal to gate movement, finding amenities and moving through security and baggage claim areas. Employees and students on large campuses and customers trying to locate stores in large shopping malls could use this technology. In a critical situation, the system can be useful in providing guidance during emergency evacuation.

## 6 Future Scope and Enhancements

The INDOORNAV paper establishes an exceptional foundation for a comprehensive navigation solution. The current system is poised for significant and exciting enhancements by integrating more features. The following are the avenues for future development:

- **Voice Navigation (NLP):** A truly hands-free experience is possible on integrating Natural Language Processing (NLP). Users would simply be able to tell the system where they want to go, and, based upon that input, the system would provide verbal directions for travel.
- **Augmented Reality (AR) Navigation:** As explored in recent research, an augmented reality-based application could provide an incredibly immersive navigation experience. By using a smartphone's camera, the application could overlay directional arrows, virtual signs, and other information directly onto the real-world view.
- **Adoption Scenarios:** Currently the algorithm has been developed taking a sample three-story building. The algorithm can be extended to many more architectural environments like hospitals which will help the user to navigate efficiently while finding wards, airports to find terminals, corporate campuses and malls, emergency evacuations etc.

Apart from these enhancements, the team is currently working on a mobile application to provide users with an interactive interface and real-time navigation access instead of static outputs.

## 7 Conclusion

Building upon the foundational work of H. A. Karimi [1] and A. Kumar and R. Sharma [5], this paper demonstrates the successful design and implementation of a fully functional, scalable indoor navigation system tailored to complex architectural environments. Using geospatial data processing and graph-based pathfinding algorithms, INDOORNAV can solve the problem of pathfinding in a large and complicated architectural building. Using open-source tools including QGIS, Flask, GeoPandas, and NetworkX and adopting the web-based accessibility principles discussed by J. Schiller and A. Voisard [21], INDOORNAV offers a cost-effective solution that requires no native application installation.

Implementing the A\* algorithm for shortest path finding, and visualising an intuitive map, the system can guarantee both accuracy and usability. The system's design choices, such as the three-tier architecture and the method of snapping room centroids to the corridor network, have proven to be robust and effective. This paper fills a significant gap in the existing complex architectural infrastructure, as no prior application has been offered dedicated for indoor navigation tailored to such environments. The system's design also lays groundwork for new innovations in the field of indoor navigation such as real-time positioning, dynamic obstacle avoidance, and other accessibility features. The application is set to evolve into a comprehensive indoor navigation platform that improves daily navigation for users.

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